

Plugging into the Future: An Exploration of Electricity Consumption Patterns using Tableau

TEAM ID	SWTID-2026-2268
PROJECT TITLE	Plugging into the Future: An Exploration of Electricity Consumption Patterns using Tableau
MAX MARKS	3 MARKS

6.3 – Data Preparation

Sheets	Type
2019 State Consumption	Map Chart
2020 State Consumption	Map Chart
Total Consumption	Vertical Bar Chart
Usage the Region	Bar Chart
Top N and Bottom N	Bar Chart
2019- and 2020-Month wise Consumption	Bar Chart
Total Consumption Region Wise	Pie Chart
Usage Before and After Lockdown	Bar Chart
Region Wise State Usage	Horizontal Bar Chart
Quarter Usage	Bar Chart
Metro City State Usage	Map Chart
Usage By Year	Line Chart

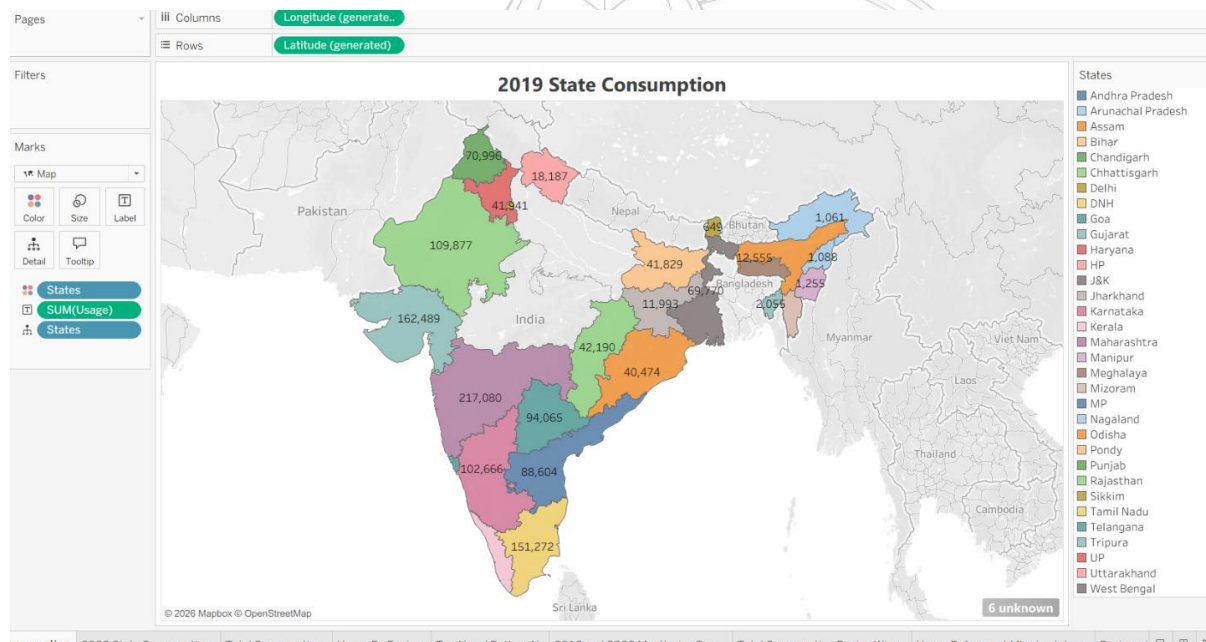
Sheet – 1 – 2019 State Consumption – Map Chart

1. In the Date Pane on the left side, right click on States.
2. Select Geographic role.
3. Choose State/Province.
4. Drag Longitude on the Columns.

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5. Drag Latitude on the Rows.
6. Drag States on Detail.
7. Drag Usage on Colors.
8. Again, Drag Usage on Label.
9. Drag Dates to Filters and Select the years 2019.
10. Click Ok.
11. Select Chart type as Map.
12. Drag States to Colors.

OUTPUT:



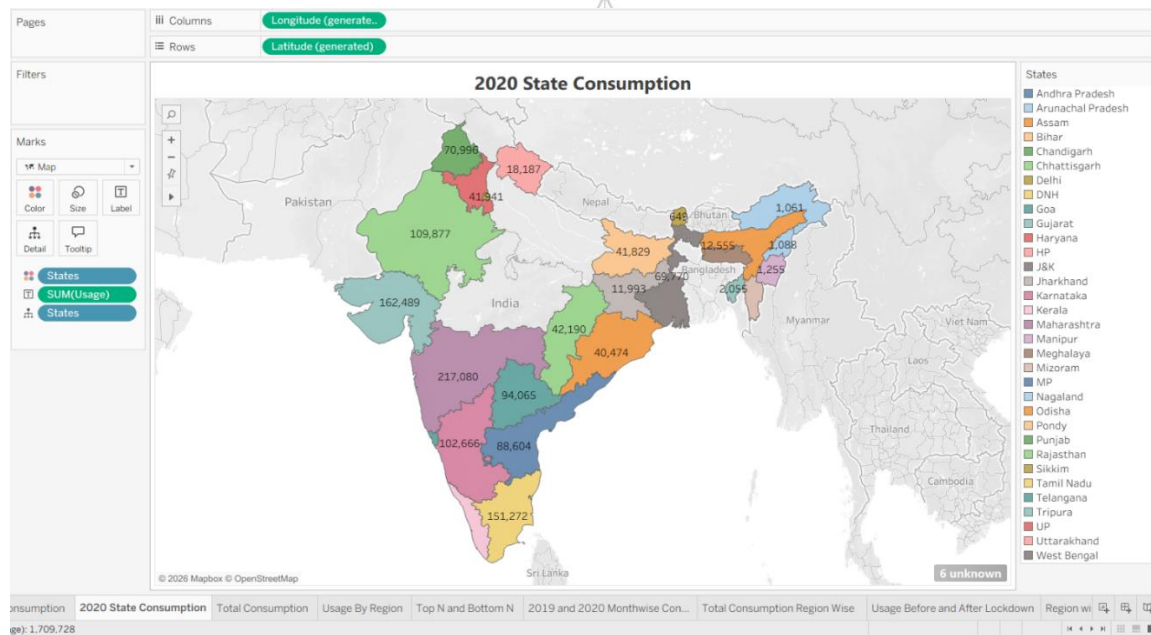
Sheet – 2 – 2020 State Consumption – Map Chart

1. In the Date Pane on the left side, right click on States.
2. Select Geographic role.
3. Choose State/Province.
4. Drag Longitude on the Columns.
5. Drag Latitude on the Rows.
6. Drag States on Detail.
7. Drag Usage on Colors.

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8. Again, Drag Usage on Label.
9. Drag Dates to Filters and Select the years 2019.
10. Click Ok.
11. Select Chart type as Map.
12. Drag States to Colors.

OUTPUT:

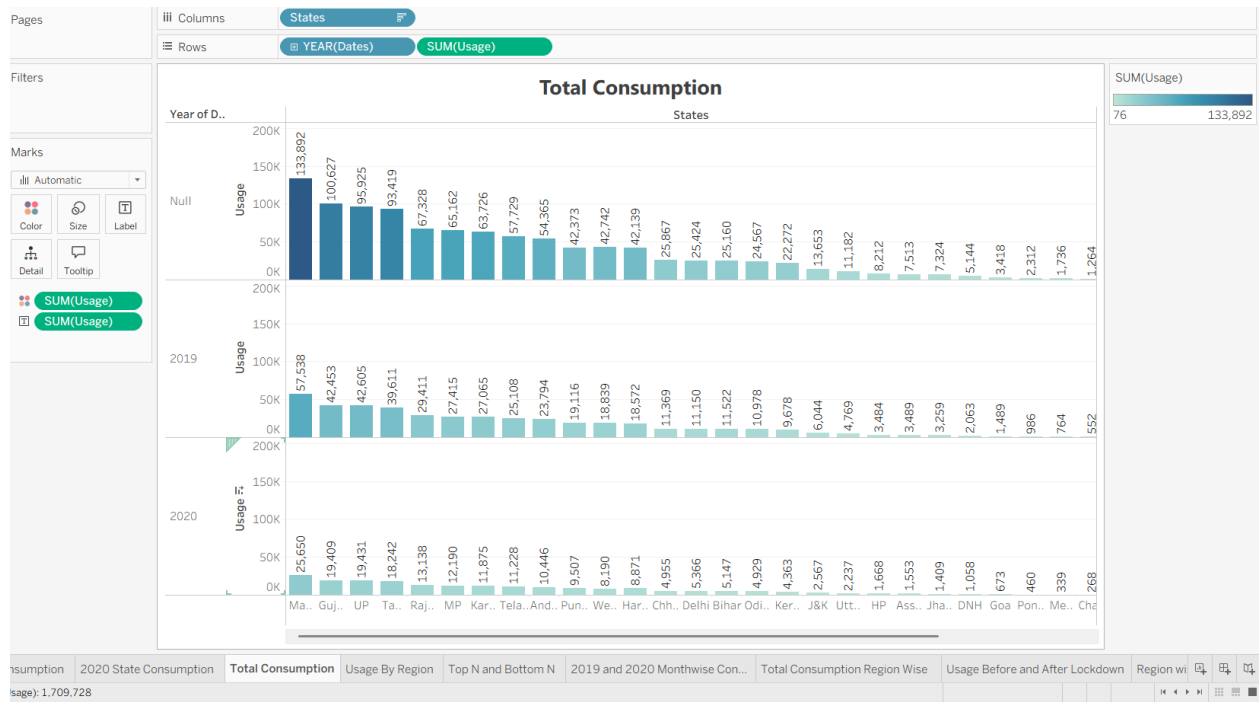


Sheet – 3 – Total Consumption – Vertical Bar Chart

1. Drag States into the rows.
2. Drag Usage into the columns.
3. Drag Dates to columns left side of the Usage.
4. Drag Sum (Usage) to Colours.
5. Drag Sum (Usage) to Label.

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OUTPUT:

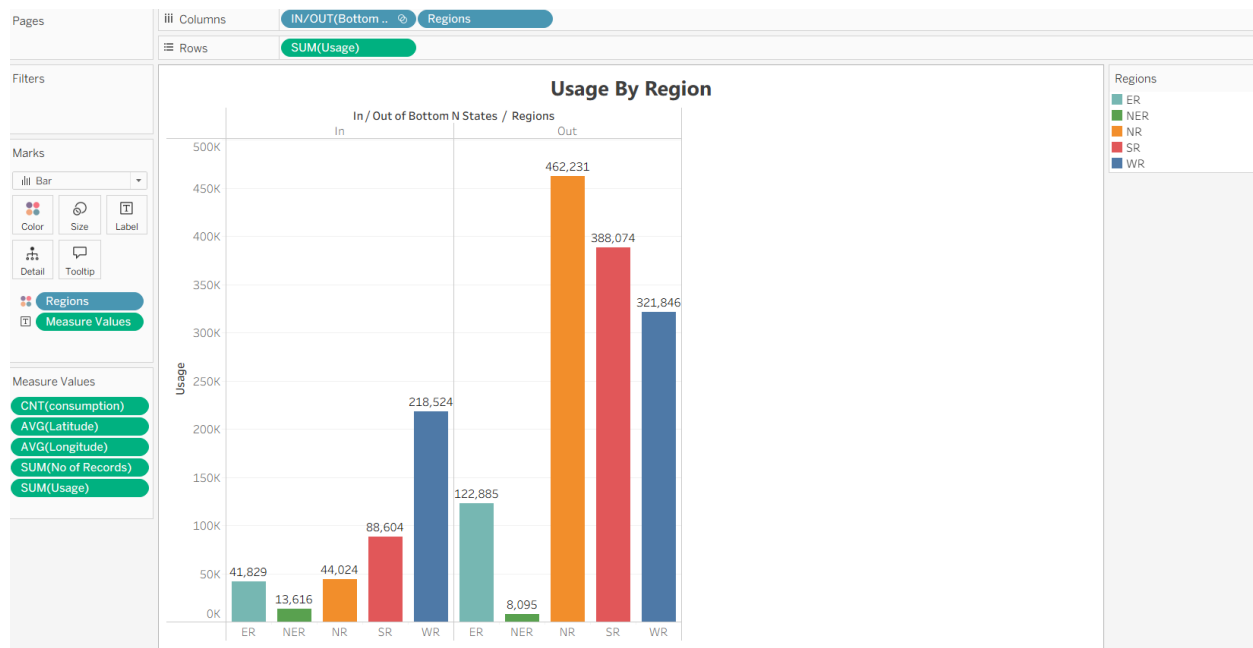


Sheet – 4 – Usage by Region – Bar Chart

1. Select the Chart type as Pie.
2. Drag Region to Colors.
3. Again, Drag Regions to Label.
4. Drag Usage to Angle.
5. Again, Drag Usage to Label.
6. Click size to increase pie size.
7. Then click on entire view.
8. In Show me Select Bar Chart.
9. Drag the Measure Values to Label.

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OUTPUT:

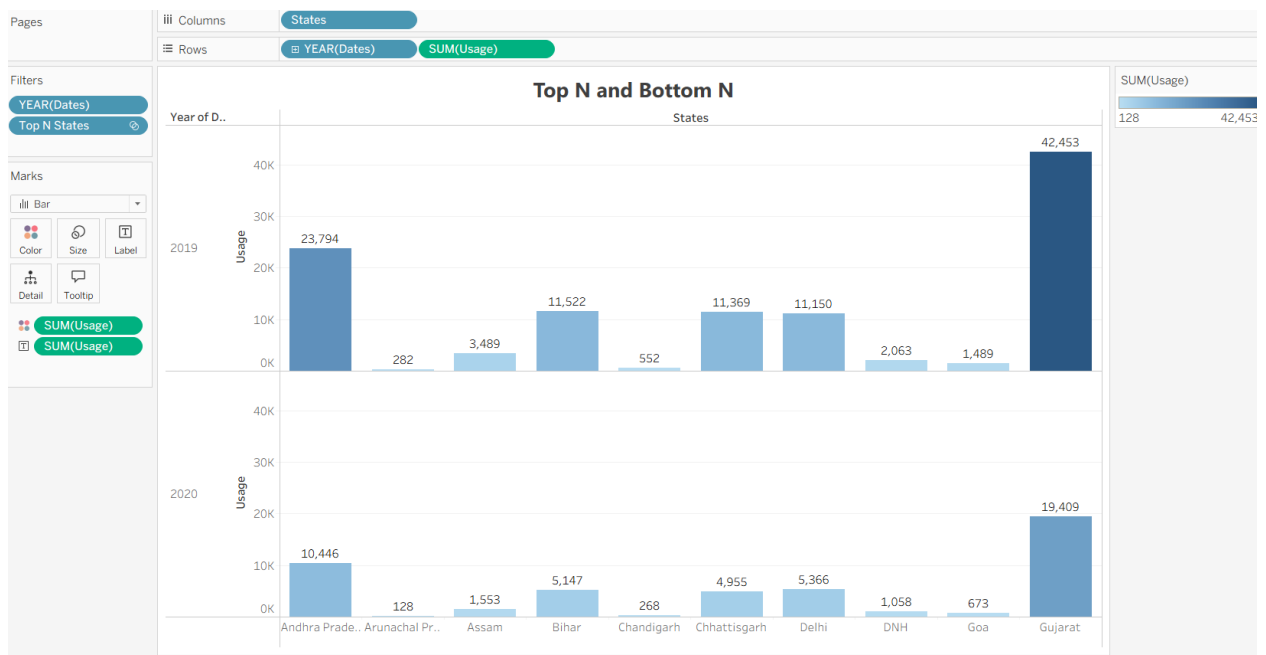


Sheet – 5 – Top N and Bottom N – Bar Chart

1. Drag States to Rows.
2. Drag Usage to Columns. Then it will automatically create a horizontal bar chart.
3. Click the sort Descending button on the toolbar.
4. Right Click on States.
5. Choose Filters.
6. Choose by field and set top = 10 sum(usage)
7. Then Click Ok.
8. Drag Usage to Label.
9. Drag Usage to Colors.
10. Then Click Entire View.

OUTPUT:

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Sheet – 6 – 2019- and 2020-Month Wise Consumption – Bar Chart

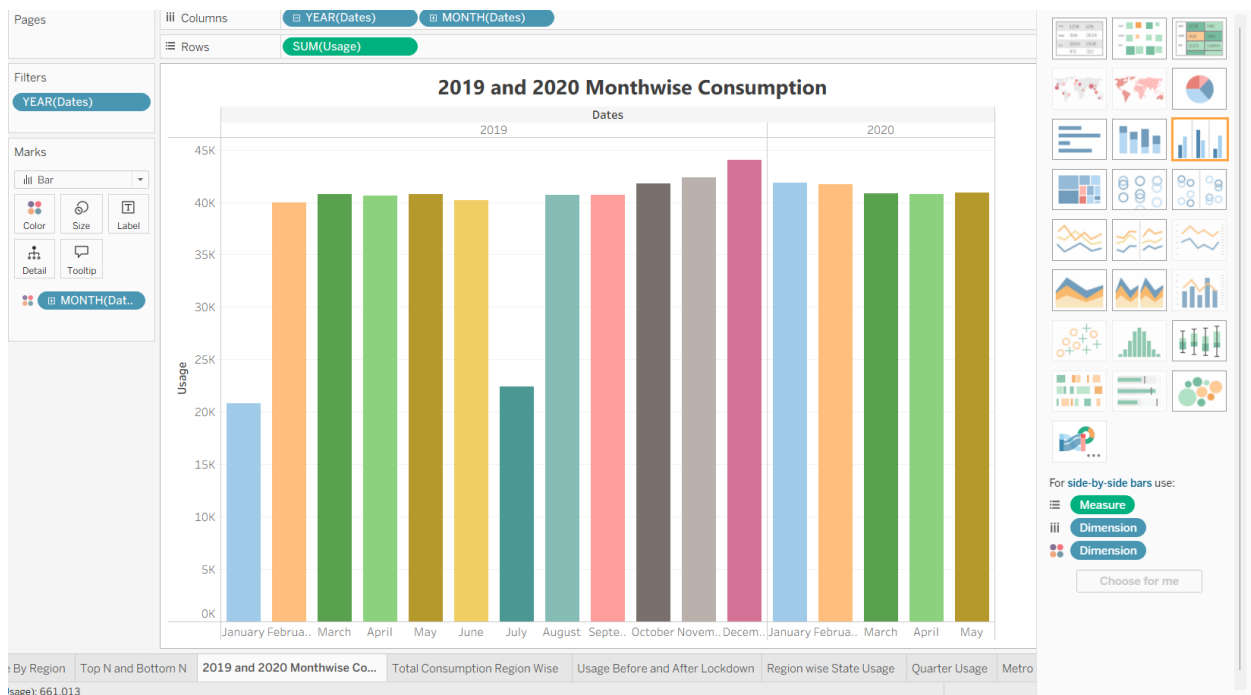
1. Drag Dates to Columns. It will automatically change year(dates).
2. Again, Drag Dates to Columns. It will change for Month.
3. Drag Usage to Rows.
4. Drag Month (Dates) to color. Press Ctrl and drag.
5. Drag year(dates) to Filters.

OUTPUT:

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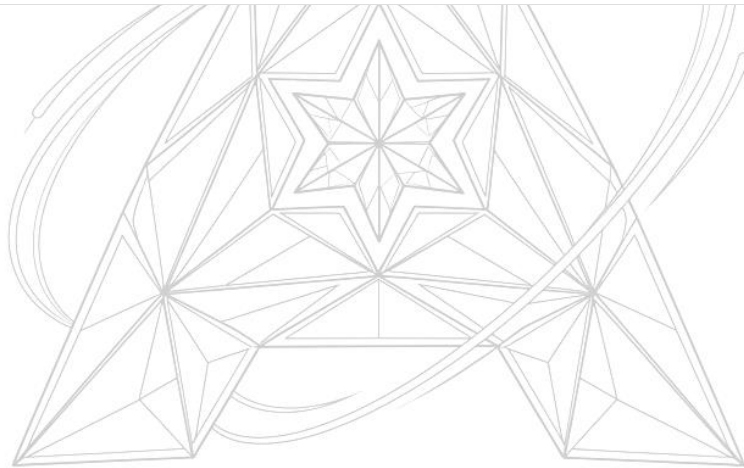
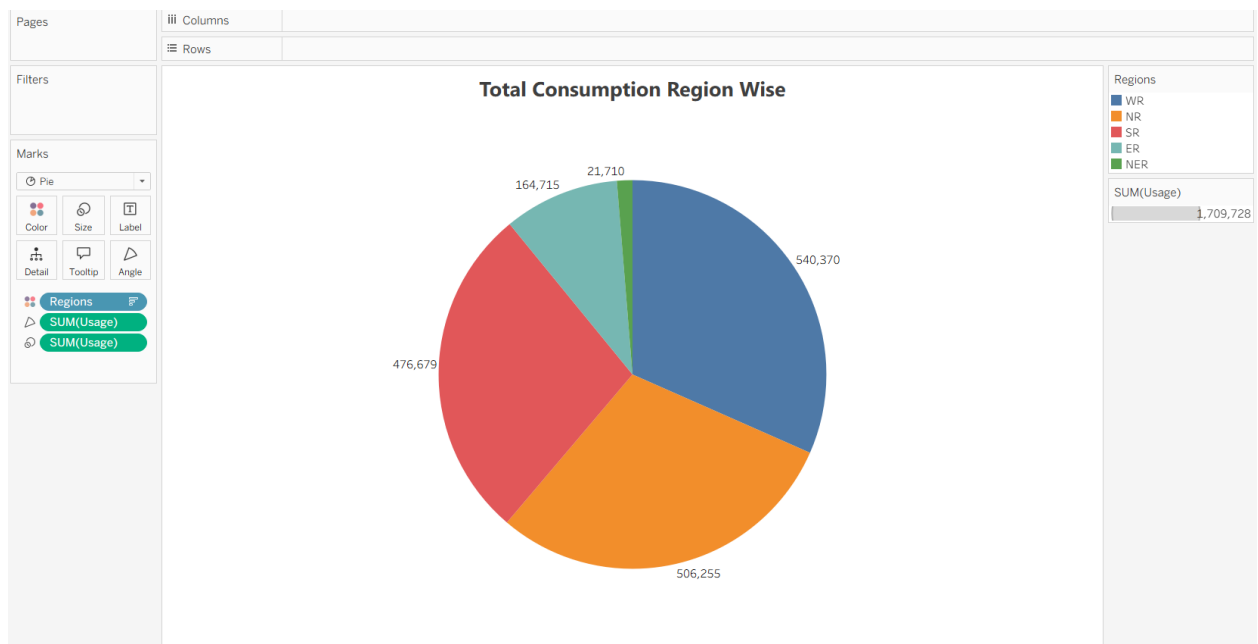


Sheet – 7 – Total Consumption Region Wise – Pie Chart

1. Drag States to the Rows.
2. Drag Usage to columns.
3. Drag Dates to columns left side of the Usage.
4. Drag Sum(usage) to colors.
5. Drag Sum(usage) to Label.

OUTPUT:

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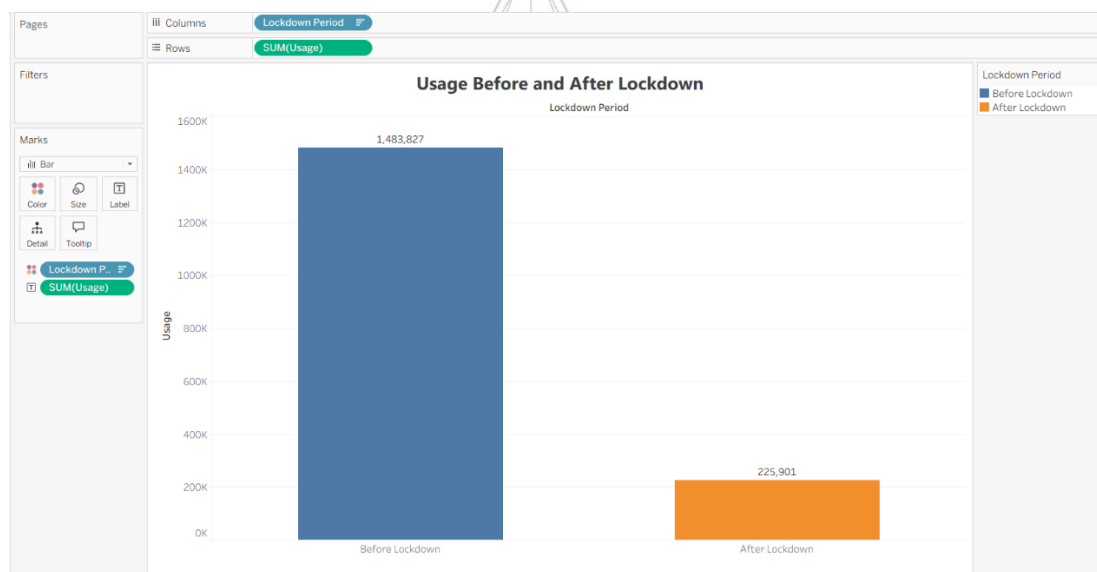
Sheet – 8 – Usage and Before and After Lockdown – Bar Chart

1. Go to the Analysis menu.
2. Create Calculated Field.
3. Name the field as Lockdown Period.
4. Enter the Formula
 - IF MONTH ([Dates]) <=3 THEN
 - “After Lockdown”
 - ELSE
 - “Before Lockdown”
 - END
5. Then Click Ok.

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6. Drag Lockdown Period to columns.
7. Drag Usage to Rows.
8. Drag Usage to Label.
9. Drag Lockdown Period to color.
10. Click Size to increase Bar size.
11. Then Select the Entire View

OUTPUT:



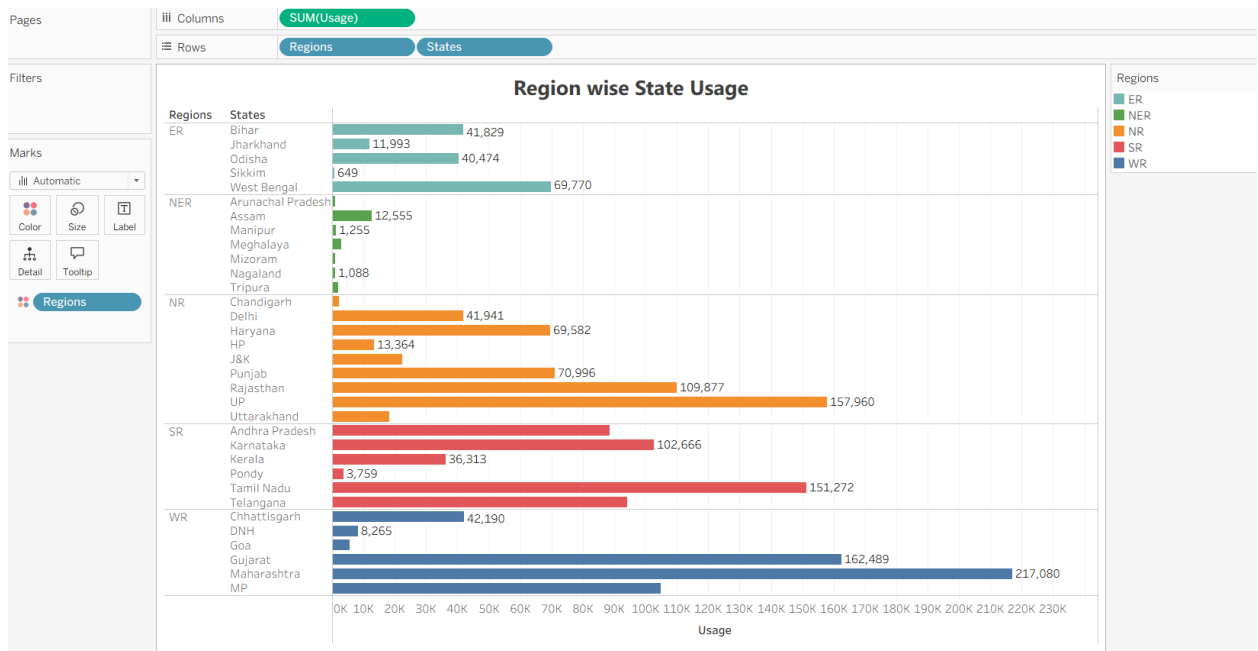
Sheet – 9 – Region Wise State Usage – Bar Chart

1. Open Show me and Select Horizontal Bar Chart.
2. Drag States to Label.
3. Drag Usage to Size.
4. Drag regions to color.
5. Then Select Entire View.

OUTPUT:



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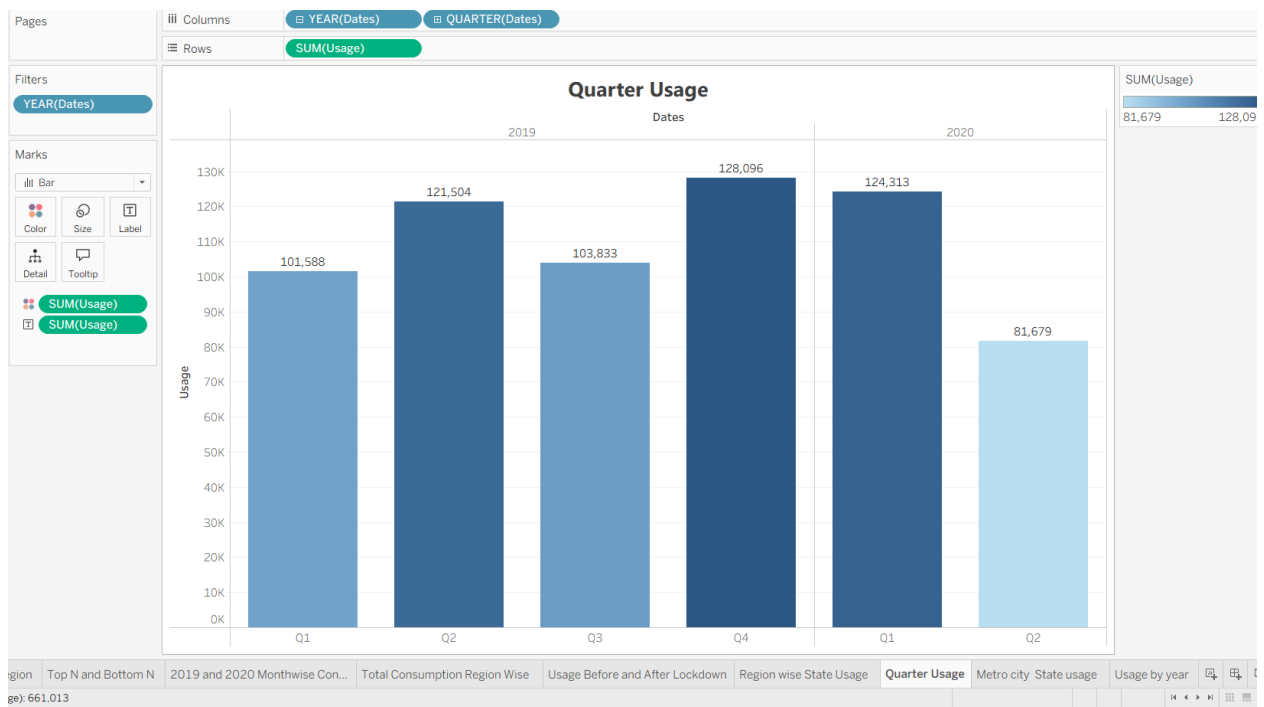


Sheet – 10 – Quarter Usage – Bar Chart

1. Drag Quarter (dates) to color. Press Ctrl to drag.
2. In the Chart type Select Bar.
3. Click Size to increase the Bar.
4. Then Click Entire View.

OUTPUT:

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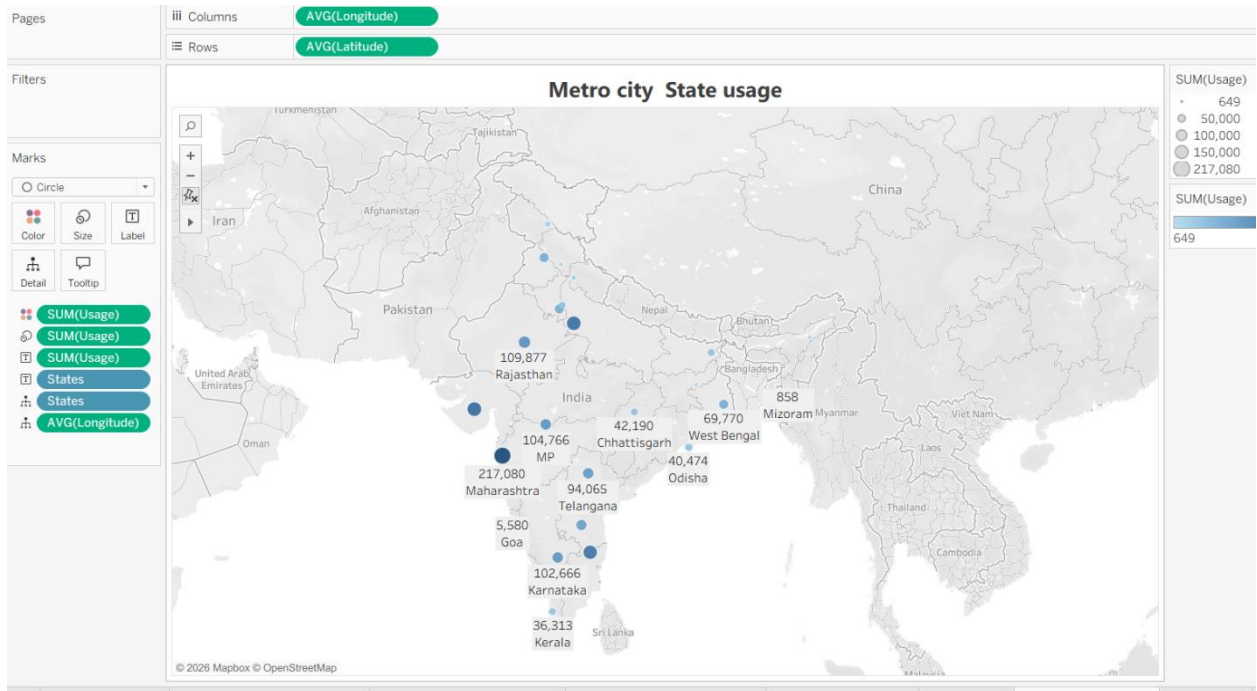


Sheet – 11 – Metro City State Usage – Map Chart

1. Drag Longitude on columns.
2. Drag Latitude on rows.
3. Drag Usage on Colors.
4. Drag Usage on Size.
5. Drag Usage on Label.
6. Drag States on Label.
7. Drag States on Detail.
8. Drag Avg (Longitude) on Detail.

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OUTPUT:



Sheet – 12 – Usage by Year – Line Chart

1. Drag Dates to columns.
2. Drag Usage to rows.
3. Drag year (dates) to colors. Press Ctrl and drag.
4. Drag usage to Label.

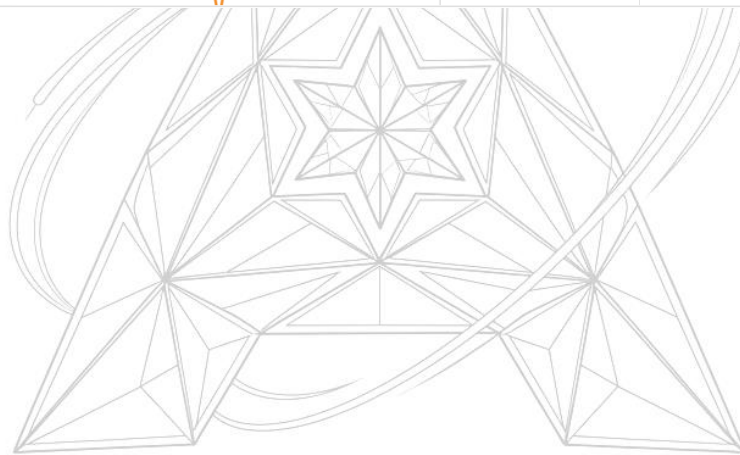
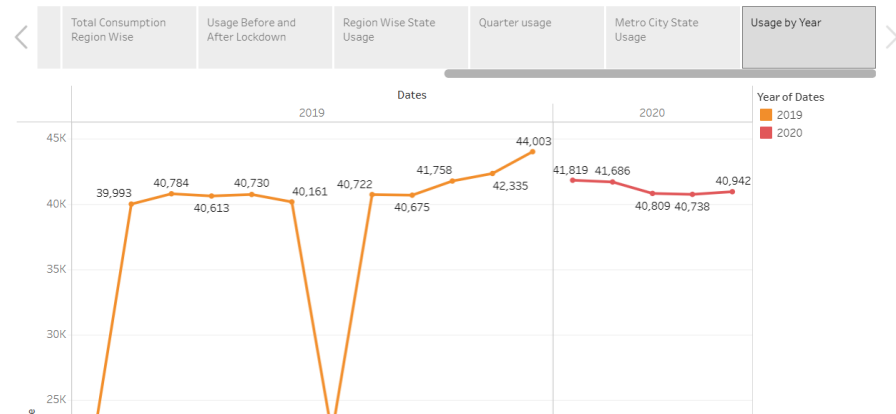
OUTPUT:

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Story

The Tableau story presents a complete view of electricity consumption patterns before and after lockdown, regional usage, quarterly analysis, metro city trends and annual consumption growth.

Story 1



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